Module		Channel Design / Hydraulic Structures
Topic	Drop structure need example	

<u>Example of need of hydraulic structures (ENCN442):</u>

A ditch to convey 1.4 m^3/s through a 7 km reach of clay type soil must be designed. The natural ground slope in the direction of the flow is 0.001. Approximately how many drop structures will be required in this reach?

- Q = 1.4 m^3/s
- 7 km of clay type soil
- natural slope 0.001

Stable channel design equations provided:

Clayey soil:

W = 3.0 $Q^{1/2}$

D = 0.75 $Q^{1/3}$

Sand soil:

W = 4.0 $Q^{1/2}$

D = 0.6 $Q^{1/3}$

PROBLEM SOLUTION:

From given equations calculate average channel width, depth, and permissible velocity:

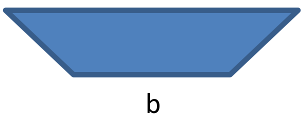
$W = 3.0 \times 1.4^{1/2} = 3.55\, m\ (average\ channel\ width)$

$D = 0.75 \times 1.4^{1/3} = 0.84\, m\ (average\ flow\ depth)$

$Q = V * A$

$V = \frac{Q}{D \cdot W} = \frac{1.4}{0.84 \times 3.55} = 0.47 \frac{m}{s}\ (permissible\ velocity\ of\ water\ in\ channel)$

Using 1.5:1 side slopes (z=1.5) – trapezoidal channel:



bed width:

b=W-zD

$b = 3.55 - 1.5 \times 0.84 = 2.29\, m$

Perimeter:

$P = b + 2 \cdot D(1 + z^2)^{\frac{1}{2}}$

Hydraulic radius:

R = DW/P

R= 0.56 m

Using Manning's

$$Q = (1/n) \cdot A \cdot R^{2/3} \cdot S^{1/2}$$

$$V = 1/n \cdot R^{2/3} \cdot S^{1/2}$$

$$S = V^2 \cdot n^2/R^{4/3}$$

Assume: n = 0.025 (smooth Earth channel)

Calculate what the permissible slope of the Channel will be using Manning's:

$$S = 0.47^2 \cdot \frac{0.025^2}{0.56^{\frac{4}{3}}} = 0.0003$$

Now calculate what the drop will be over the 7 km:

7 km x 0.001 = 7 m of drop

Permissible drop in a channel bed level at 0.0003 will be 7000 m X 0.0003 = 2.1 m

Drop difference:

7 m – 2.1 m = 4.9 m (drop needed to ensure a permissible channel)

In other words, you need 4.9 m of drop structures in your channel to ensure the stability of the channel.   To reduce excavation costs you could choose to implement 5 drop structures evenly distributed along your 7 km of channel each of about 1 m of drop.